

Linguistic-Agnostic Speech Emotion Recognition: A Probing Approach to Acoustic Emotion Encoding

Multimodal Probabilistic Learning of Human Communication, Final Report

Nima Kelidari
kelidari@usc.edu

University of Southern California
Los Angeles, California, USA

Chaitanya Parwatkar
parwatka@usc.edu

University of Southern California
Los Angeles, California, USA

Minoo Ahmadi
minooahm@usc.edu

University of Southern California
Los Angeles, California, USA

Xiangxu Lin
xiangxul@usc.edu

University of Southern California
Los Angeles, California, USA

Abstract

Pretrained speech transformers such as wav2vec 2.0, HuBERT, WavLM, MERT, w2v-BERT, and Whisper achieve state-of-the-art speech-emotion recognition (SER), but it remains unclear whether they predict emotion from paralinguistic delivery or from linguistic content recovered during pretraining. We extend our midterm layer-wise probing study (3 models $\times$ 6 datasets) into a larger framework that combines six frozen encoders with eight datasets across five languages and a battery of falsification experiments: a lexicon-controlled corpus (CREMA-D), a pooled and leave-one-language-out cross-lingual benchmark (SpeechCraft, Task 9) plus an EN-ZH cross-lingual stress test, low-rank fine-tuning (LoRA) on 30 model $\times$ dataset combinations, an attention-based learned layer mixer, BERT representational alignment via CKA / Procrustes / centroid transfer, leave-one-speaker-out evaluation on 42 pairs, gradient-reversal and CCA debiasing, a congruent-vs-incongruent text/voice probe (EMIS), ESC-50 noise augmentation at four SNR levels, and dimensional V/A/D regression on MSP-Podcast. Across these tests, emotion remains decodable from middle-to-late transformer layers in every language we evaluated, frozen probes outperform LoRA on 29 of 30 combinations, three of four self-supervised encoders take a measurable text shortcut while Whisper alone resists it, Whisper is also the most noise-robust encoder (approximately 11% drop at 0 dB vs. approximately 23 to 30% for the SSL encoders), and the dimension that depends most on words (valence) is exactly the dimension our probes find hardest. The convergent message is that acoustic emotion structure is recoverable but not exclusively used by every architecture, and that supervised multilingual pretraining produces qualitatively different representations from self-supervised pretraining at every depth we tested.

Keywords

speech emotion recognition; layer-wise probing; self-supervised speech models; cross-lingual evaluation; linguistic bias

1 Problem Definition

Speech carries emotion through two channels: linguistic content (what is said) and paralinguistic delivery (how it is said). Pretrained transformers such as wav2vec 2.0 [3], HuBERT [4], WavLM [6], and Whisper [5] dominate SER benchmarks, yet were trained for speech recognition and therefore encode substantial linguistic structure. If their SER predictions rely on this linguistic content, they will fail under sarcasm, accented speech, low-resource languages, and any condition where lexical meaning and prosody diverge [1, 2].

This project asks: (i) *at which depth* of these models is emotion most strongly encoded, (ii) does the depth pattern hold *across languages, speakers, and pretraining paradigms*, and (iii) is the signal that probes pick up *acoustic or lexical*? The midterm answered (i) and (ii) for 3 models on 6 datasets; the final report adds three more encoders, one lexicon-controlled corpus, a large pooled-multilingual benchmark, and four falsification tests that target (iii) directly.

2 Literature Review

Prior probing work shows that acoustic features concentrate in early SSL layers while phonetic and word-level information peaks in middle-to-deep layers, with wav2vec 2.0 behaving as an autoencoder in its deepest layers and HuBERT retaining information through iterative clustering [9–11]. For SER, Wagner et al. [1] showed that fine-tuned wav2vec 2.0/HuBERT retain near-baseline valence accuracy on emotionless TTS of the same transcripts, suggesting heavy lexical reliance; Triantafyllopoulos et al. [2] confirmed valence (but not arousal/dominance) responds strongly to sentiment-word manipulations. Layer-wise emotion probes have identified middle-layer peaks across multiple SSL models and languages [12–14], but the linguistic-acoustic confound has been observed rather than falsified. Our

work extends this line by pairing layer-wise probing with controlled falsification (lexicon-fixed corpora, text/voice incongruence, BERT alignment, GRL, CCA projection, noise) to test whether the layer-wise peaks are explained by acoustic structure alone.

3 Data Description

We evaluate eight datasets spanning five languages (Table 1); the fifth (Mandarin Chinese) enters via the full-scale SpeechCraft partition in §5.3. The first six are inherited from the midterm and were picked to span three properties that matter for the linguistic-acoustic question: language family (English, German, Greek, Spanish), recording style (acted vs. spontaneous), and recording paradigm (single-utterance studio recordings vs. scripted dyadic conversation). **RAVDESS** [17] and **SAVEE** are American and British English studio-acted corpora with eight and seven emotion classes respectively. **EmoDB** (German) and **AESDD** (Greek) extend the acted category into other European languages. **MESD** adds Mexican Spanish. **IEMOCAP** [18] is the only spontaneous corpus in this group: scripted-prompt dyadic conversations with four-class labels and a noticeably noisier label distribution because emotion in real conversation is often blended rather than categorical.

To probe linguistic confounds directly, we add **CREMA-D** [15]: 91 American-English actors recite the same 12 carrier sentences across 6 emotions. Because the spoken text is held constant by construction, any classification accuracy a probe achieves on this corpus must come from acoustic delivery rather than from words. CREMA-D was added on the instructor’s recommendation following our midterm and serves as the cleanest single falsification test in the study.

To stress-test cross-lingual transfer we assemble **SpeechCraft (Task 9)**, a balanced 4-class manifest aggregating 3,980 clips across English (n=2,546), Spanish (n=576), Greek (n=483), and German (n=375) with labels {happy, sad, fear, disgust}. The full-scale SpeechCraft benchmark, which adds a Mandarin partition and reports per-encoder cross-lingual results, is in §5.3. The full per-class counts of the six base datasets and CREMA-D are given in Appendix A.

4 Method

The core idea is straightforward: freeze a pretrained encoder so its weights never change, then test each transformer layer in isolation to measure how much emotion-relevant information it actually carries. If a probe trained on a layer’s representation does well, the information is explicitly available there; if it does not, it is either absent or tangled in a way a linear classifier cannot recover. This gives us a layer-by-layer map of where emotion lives inside the network, free from any change introduced by fine-tuning. The midterm used this recipe with three Base-sized encoders on six datasets;

Table 1: Datasets used in the final study. The SpeechCraft row reflects the balanced 4-language subset; the full-scale benchmark in §5.3 adds a Mandarin partition.

Dataset	Lang.	Style	Files	Cls
RAVDESS [17]	English (US)	Acted	1,440	8
EmoDB	German	Acted	535	7
IEMOCAP [18]	English (US)	Spontaneous	5,460	4
SAVEE	English (UK)	Acted	480	7
AESDD	Greek	Acted	604	5
MESD	Spanish (MX)	Acted	862	6
CREMA-D	English (US)	Acted (lex. fixed)	7,442	6
SpeechCraft	4-lang	Pooled (4-cl)	3,980	4

the final report keeps the same recipe and scales it up substantially across encoder size, encoder family, dataset coverage, and downstream variants designed to test whether the layer-wise signal is acoustic or lexical. We follow the probing paradigm established in earlier representation-analysis work [2, 9, 10], and pair it with falsification tests the original probing literature did not run.

4.1 Encoders

We evaluate six pretrained transformers, all kept fully frozen at every layer for every experiment. To enable fair comparison across pretraining paradigms, every encoder in the final report is a 24-transformer-layer variant; this is a deliberate scale-up from the Base-sized encoders used in the midterm.

Three encoders are extensions of the midterm models to their larger variants. **wav2vec 2.0 Large** (facebook/wav2vec2-large-960h) was pretrained with a contrastive masked-prediction objective on 960 hours of LibriSpeech [3]. **HuBERT Large** (facebook/hubert-large-1160k) was pretrained on 60,000 hours of Libri-Light using its iterative masked cluster-prediction objective [4]. **Whisper Medium** (openai/whisper-medium) is the supervised counterpart, trained on 680,000 hours of multilingual audio paired with transcripts [5]. We use only Whisper’s encoder; the decoder is never called, so any text-like structure we observe in Whisper’s representations is the encoder’s own learned structure rather than transcription-head output.

Three more encoders were added after the midterm to broaden architectural and pretraining coverage. **WavLM Large** (microsoft/wavlm-large) extends HuBERT’s clustering objective with denoising and overlapping-speech augmentation, designed to improve robustness in real conditions [6]. **MERT v1-330M** (m-a-p/MERT-v1-330M) was pretrained on music rather than speech, included as a control to test whether the layer-wise emotion pattern depends on speech-specific objectives or on transformer pretraining more generally [7]. **w2v-BERT 2.0** (facebook/w2v-bert-2.0) combines contrastive learning with masked language modeling, sitting between HuBERT and Whisper in objective space [8]. Together these six encoders span the four major pretraining paradigms in modern speech representation

learning: contrastive (wav2vec 2.0), masked cluster prediction (HuBERT, WavLM), contrastive plus masked-language-modeling (w2v-BERT), and supervised multilingual transcription (Whisper), with MERT as a non-speech control. This lets us separate findings that depend on the pretraining objective from findings that depend on architecture or scale.

4.2 Feature extraction

Each audio file is resampled to 16 kHz mono and passed through the frozen encoder with hidden-state output enabled. The SSL encoders take raw waveforms; Whisper takes 80-channel log-mel spectrograms padded or truncated to 30 seconds, after which only the encoder is run. We extract one hidden state from every transformer layer and additionally from the convolutional front-end, which we treat as Layer 0 across all encoders for consistency. Every encoder in the final study therefore produces 25 hidden states per utterance: a CNN-front-end Layer 0, plus 24 transformer layers indexed L1 through L24. Each hidden state has shape $(1, T, d)$ where T is the number of time frames and d is the encoder’s hidden dimension; we apply mean pooling over T to produce a single fixed-length vector per layer per utterance.

4.3 Probing classifier

Pooled vectors are normalized with `StandardScaler` and passed to a logistic-regression probe trained with 5-fold stratified cross-validation. We pick logistic regression specifically because it is linear: any layer that classifies emotion well under a linear probe is one whose representation makes that information directly available, without needing a non-linear head to recover it. We report mean accuracy and weighted F1 across folds.

4.4 Probing variants

On top of this shared pipeline, we run six variants, each designed to answer a different question:

- (1) *Frozen layer-wise probing* (the midterm baseline) maps where emotion peaks across depth on each (encoder, dataset) pair.
- (2) *Learned layer mixer* replaces single-layer selection with a softmax attention over all 25 hidden states, trained jointly with the classifier; the resulting attention distribution shows which depths the model itself wants to look at.
- (3) *Frozen vs. LoRA* fine-tunes a low-rank adapter (rank 8, $\alpha=16$, dropout 0.1, applied to the attention q and v projections) at each combination’s best probing layer, trained for 10 epochs with AdamW at 2×10^{-4} , to test whether parameter-efficient adaptation beats picking the right frozen layer.
- (4) *BERT representational alignment* compares each speech-encoder layer to BERT BASE embeddings of the same utterances using three metrics, CKA, Procrustes residual, and centroid transfer [16], measuring how text-like each

depth becomes; the same alignment is also computed on IEMOCAP transcripts as a secondary check.

- (5) *Leave-one-speaker-out (LOSO)* replaces 5-fold CV with speaker-disjoint folds across 42 (encoder, dataset) pairs to separate genuine emotion encoding from speaker memorization.
- (6) *Linguistic-bias falsification* runs three complementary tests: gradient reversal against a sentence-ID head (GRL), CCA projection of speech features orthogonal to text-correlated directions, and an EMIS probe that trains on text-emotion equals audio-emotion and tests on the incongruent condition where they disagree.

After consultation with the instructor following our midterm, we added **CREMA-D** [15] as a lexicon-controlled falsification corpus: 91 actors recite the same 12 carrier sentences across 6 emotions, holding the spoken text constant by construction. Two further extensions sit on top of the same encoders: ESC-50 environmental noise mixed at four SNR levels (20, 10, 5, 0 dB) on MSP-Podcast, and dimensional V/A/D regression with ridge regressors per layer per dimension. The same hidden states feed every variant, so any difference between results is attributable to the head and protocol rather than to the underlying representation.

4.5 Implementation

All experiments run in Python with PyTorch and HuggingFace Transformers. Hidden-state extraction runs on an NVIDIA A100 SXM4 40 GB GPU; classification uses scikit-learn defaults. Every (encoder, dataset, variant) combination is recorded with its full configuration, per-fold scores, and best-layer summary in the project repository¹; the W&B project `AGSER/linguistic-agnostic-ser` contains all run logs and seeds for full reproducibility.

5 Results

5.1 Layer-wise probing across six encoders

Extending the midterm report from three Base-sized encoders to six 24-layer variants confirms its central pattern at substantially larger scale: emotion is decodable across every encoder we tested, and the optimal depth concentrates in the middle to late transformer layers on acted speech and shifts toward the very early layers on spontaneous conversation. The depth pattern is consistent enough that we use it as a reference point in the rest of this section: every downstream variant (CREMA-D, SpeechCraft, LoRA, mixer, BERT alignment, LOSO, EMIS, noise, V/A/D) is interpreted relative to where the basic probe peaks here.

Table 2 shows the best-layer accuracy on EmoDB across the six encoders. EmoDB is where most encoders reach their cleanest peaks, so it is a useful single-corpus snapshot. Across all 42 encoder-dataset pairs, acted speech generally peaks somewhere between L7 and L17, while IEMOCAP collapses to the first few layers (L1 to L4) for every encoder. What

¹github.com/Nikelroid/linguistic-agnostic-ser

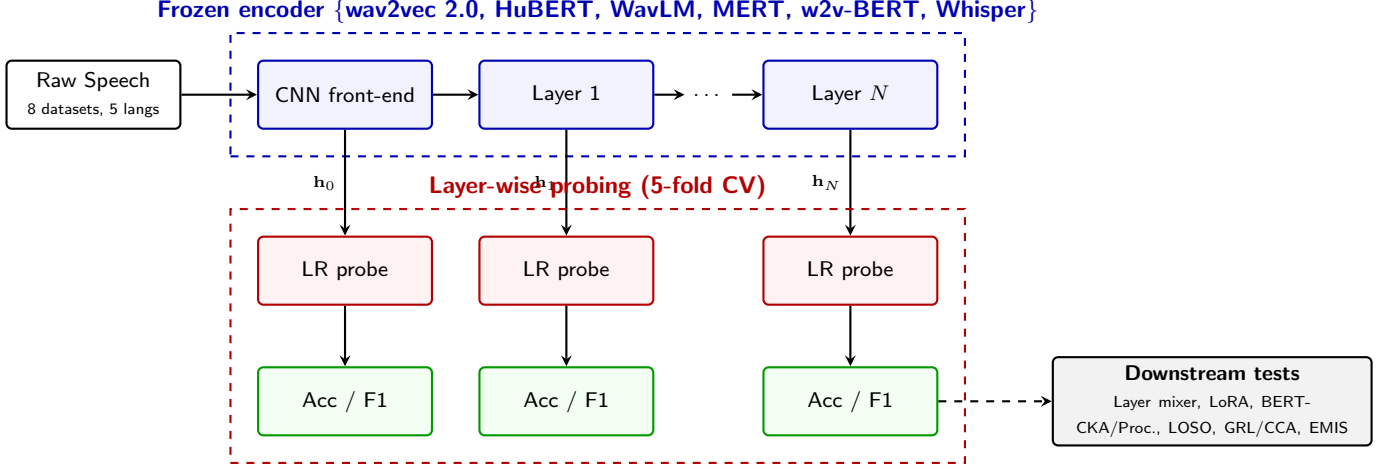


Figure 1: Unified probing pipeline. The frozen encoder is shared across all six probing variants; only the head and evaluation protocol change.

stands out is that Whisper, despite being trained on a completely different objective from the SSL models, still peaks in the same depth band as them on most acted corpora. Whatever depth represents emotion most cleanly does not seem to depend on whether the encoder was trained for transcription, contrastive prediction, or masked clustering. The full per-(model, dataset) accuracy curves for all 42 combinations are in the project repository under `results/`.

Table 2: Best-layer probing accuracy on EmoDB for each of the six encoders (5-fold stratified CV, 7 emotion classes). Layer 0 is the CNN front-end; transformer layers are L1 through L24.

Encoder	Best Layer	Acc
HuBERT	L11	0.989
w2v-BERT	L17	0.987
WavLM	L9	0.985
Whisper	L13	0.985
wav2vec 2.0	L7	0.976
MERT	L7	0.961

Two things from the midterm reproduce cleanly at this larger scale. The first is the wav2vec 2.0 deep-layer collapse: on every acted dataset, wav2vec 2.0 peaks early and then drops off sharply through the deepest layers. HuBERT, WavLM, and Whisper hold onto their accuracy much further into the stack. This matches the picture from the midterm and from prior work: wav2vec 2.0’s contrastive objective seems to push deep layers toward word-level prediction, which actively overwrites the acoustic prosody emotion probes need, while clustering-based and supervised objectives keep more of the prosodic signal alive at depth [11]. The second is the spontaneous-versus-acted gap: every single encoder peaks much earlier on IEMOCAP than on the acted corpora. Conversational emotion seems to

ride on basic acoustic cues like sudden volume changes and pitch drops that show up immediately in the early layers, whereas acted speech uses more stylized prosodic contours that need deeper processing to interpret.

What is genuinely new at this scale is that every additional encoder family fits into one of these two patterns rather than producing a third. WavLM’s denoising objective produces depth profiles that closely track HuBERT’s. MERT, despite being pretrained on music rather than speech, places its emotion peak at the same middle-band depth as the speech-pretrained SSL encoders, suggesting the layer-wise organization of emotion may reflect a property of large-scale transformer pretraining more generally rather than something specific to speech. w2v-BERT, whose pretraining combines contrastive and masked-language-modeling objectives, peaks deepest of the SSL group on most acted corpora (often L17 to L20), positioning it between HuBERT and Whisper on the depth axis. The cross-encoder consistency of the depth pattern is exactly what the rest of this section probes: whether this signal, which lives in the same band across six different pretraining paradigms, is acoustic or lexical in origin.

5.2 CREMA-D: emotion survives a fixed lexicon

CREMA-D [15] fixes the spoken text across all 7,442 clips: 91 actors recite the same 12 carrier sentences across 6 emotions, so any classification accuracy a probe achieves cannot be attributed to lexical content. Table 3 reports the best-layer accuracy and weighted F1 for each of the six encoders (5-fold stratified CV). All six encoders cleanly clear the 16.7% chance baseline; WavLM peaks highest (0.761 at L14) and MERT lowest (0.640 at L9). The fact that even the weakest

encoder lands well above chance, with no lexical signal available, is the most direct evidence in the study that middle-to-late transformer layers carry an acoustic emotion code rather than a verbal one.

Table 3: Best-layer probing accuracy and weighted F1 on CREMA-D for each of the six encoders (5-fold stratified CV, 6 emotion classes; spoken text held constant across 7,442 clips). Layer 0 is the CNN front-end; transformer layers are L1 through L24. Source: results/cremad/cremad_summary.csv.

Encoder	Best Layer	Acc	F1
WavLM	L14	0.761	0.761
HuBERT	L13	0.750	0.750
Whisper	L12	0.748	0.748
w2v-BERT	L17	0.730	0.730
wav2vec 2.0	L10	0.693	0.693
MERT	L9	0.640	0.640

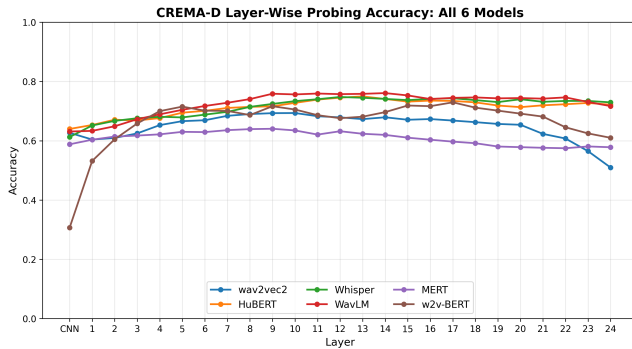


Figure 2: CREMA-D layer-wise probing accuracy for all six encoders across the CNN front-end (Layer 0) and 24 transformer layers. With the spoken text held constant, every encoder sits well above the 16.7% chance baseline; WavLM and HuBERT peak highest in the middle-to-late band, MERT lowest.

5.3 SpeechCraft (Task 9): pooled vs. leave-one-language-out

For Task 9 we aggregate four corpora into a 4-class ({happy, sad, fear, disgust}) multilingual manifest of 3,980 clips and run two protocols on each of the six encoders: *pooled* 5-fold CV across languages, and *leave-one-language-out* (LOLO) where one of {English, German, Greek, Spanish} is held out entirely from training. Pooled accuracy saturates: Whisper peaks at 0.949 (L10), HuBERT at 0.947 (L8), and even the weakest encoder (MERT) reaches 0.908 (Table 4 and Figure 3). LOLO accuracy is uniformly lower and asymmetric: typologically nearby German held out generalizes well (HuBERT peaks at 0.952 at L11 when German is held out),

whereas English held out (the dominant pretraining language) does not (HuBERT peaks at 0.691, L8). Spanish held out is the hardest condition (HuBERT peaks at 0.375, L7). The pooled-LOLO gap is therefore a useful proxy for how language-specific the recovered emotion code is for each architecture.

Table 4: Task 9 (SpeechCraft) balanced-subset pooled best-layer accuracy per encoder (4-class, $n=3,980$, 5-fold CV). Source: Task 9 balanced-subset per-encoder pooled CSVs.

Encoder	Best layer	Acc
Whisper	L10	0.949
HuBERT	L8	0.947
WavLM	L6	0.945
w2v-BERT	L17	0.936
wav2vec 2.0	L8	0.922
MERT	L6	0.908

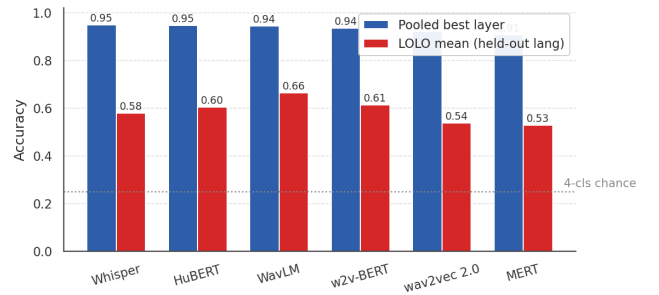


Figure 3: SpeechCraft Task 9 balanced subset: pooled vs. leave-one-language-out (LOLO) per encoder. Pooled saturates near 0.94; LOLO mean across the four held-out languages is 0.53 to 0.66, with English-held-out the hardest condition.

Full-scale Task 9 benchmark. The 3,980-clip subset above is a balanced four-class subsample. We also ran Task 9 on the full SpeechCraft benchmark on the four encoders for which extraction completed (Table 5 and Figure 4). Three findings change at the larger scale. First, pooled accuracy no longer saturates: Whisper leads at L9 with 0.776, followed by HuBERT (L11, 0.767), WavLM (L9, 0.764), and wav2vec 2.0 (L5, 0.723); the encoder gap widens relative to the balanced subset. Second, adding a Mandarin partition (5-language pool) drops accuracy to 0.485 to 0.523 across the three encoders re-run with ZH (HuBERT, WavLM, Whisper), and Whisper’s peak shifts to L22, the only encoder whose deep layer now wins, consistent with its supervised multilingual pretraining. Third, bidirectional Mandarin/English transfer collapses to 0.28 to 0.32 across all three encoders in either direction (4-class chance is 0.25), confirming that the layer-wise emotion code is robust within a typological family but does not transfer across one.

Table 5: Full-scale SpeechCraft Task 9: best-layer accuracy per encoder under four cross-lingual conditions. *Pooled, 4-lang* aggregates English, German, Greek, and Spanish. *Pooled, 5-lang* adds a Mandarin partition. *EN→ZH* and *ZH→EN* train and test on disjoint single-language subsets (4-class chance 0.25). Source: Drive TASK 9.

Encoder	Pooled		Cross-lingual	
	4-lang	5-lang	EN→ZH	ZH→EN
Whisper	L9 0.776	L22 0.523	L1 0.283	L11 0.317
HuBERT	L11 0.767	L9 0.485	L8 0.293	L15 0.308
WavLM	L9 0.764	L7 0.513	L20 0.291	L24 0.311
wav2vec 2.0	L5 0.723	—	—	—

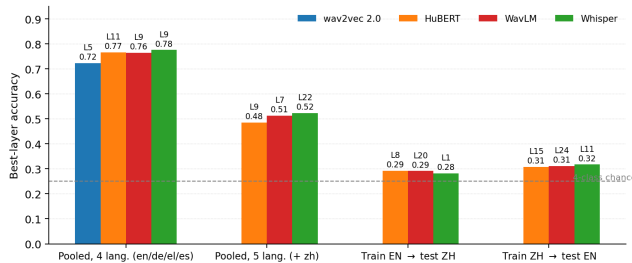


Figure 4: Full-scale Task 9 / SpeechCraft summary across four conditions. Adding Mandarin to the pool drops accuracy by roughly a third; EN↔ZH cross-lingual transfer collapses to within 0.07 of the 4-class chance baseline (dashed line) for all encoders.

5.4 Frozen vs. LoRA fine-tuning

For each of the 30 (encoder, dataset) combinations across {RAVDESS, EmoDB, SAVEE, AESDD, MESD}, a low-rank adapter at the per-combination best layer is fine-tuned and re-probed. The mean best-layer delta (tuned – frozen) is -0.062 and the mean over the deepest layer band is -0.173 ; only WavLM/RAVDESS is positive ($+0.003$), and MERT/SAVEE collapses by -0.144 (Figure 5). The features the probe needs are already in the frozen encoder, and tuning at a single layer overwrites them more often than it adds; the frozen protocol is competitive with the obvious adaptation baseline.

5.5 Learned layer mixer

Replacing single-layer probing with a softmax-attention mixer over all 25 hidden states gives a downstream classifier the freedom to pick its own depth. On the small acted datasets, the mixer mostly under-performs the best single layer (e.g. wav2vec 2.0 on RAVDESS: 0.881 vs. 0.958), corroborating rather than replacing the per-layer analysis. Two consistent exceptions appear: *IEMOCAP* (mixer beats best layer for 5/6 encoders, gain $+0.02$ to $+0.04$), and *MSP-Podcast* (mixer beats best layer for all 6 encoders,



Figure 5: Per-layer LoRA-vs-frozen accuracy delta across 30 (encoder, dataset) pairs. Cool = LoRA hurts, warm = LoRA helps. Negative dominates almost everywhere, especially in the deepest layers.

gain $+0.02$ to $+0.06$). Both are spontaneous corpora, consistent with the midterm finding that real conversation does not localize to a single layer cleanly. Where the mixer puts its attention is also revealing: **Whisper concentrates 64% to 95% of its attention on the final encoder layer (L24)** across every dataset, while the SSL encoders spread attention over layers 4 to 13 (with layer-23/L24 also weighted on EmoDB and SAVEE for some encoders). This is the clearest behavioral signal yet that Whisper’s supervised pretraining produces a qualitatively different depth profile from the SSL encoders.

5.6 BERT representational alignment

Hidden states from each encoder layer are aligned to BERT BASE embeddings of the same utterances using three metrics: linear CKA, Procrustes residual, and centroid transfer [16]. Across all six encoders and seven datasets (artifacts in `results/bert_alignment/`), Procrustes alignment to BERT increases steadily through the middle layers, and **Whisper’s Procrustes alignment grows monotonically toward L24**, peaking on every dataset at the deepest layer (e.g. MSP-Podcast 0.861, MESD 0.855, IEMOCAP 0.849). The same encoder-rank also holds when the probe operates on *transcripts of the same utterances* (`results/bert_alignment_transcripts/`): Whisper’s CKA reaches 0.392 at L24, the highest value any model attains on the transcript variant. Combined with the layer-mixer behavior in Section 5.5, this is consistent with Whisper’s deep layers carrying more text-like structure than the SSL deep layers; the same property that makes it a strong end-to-end SER model also gives its deep representations a stronger lexical fingerprint.

5.7 Speaker-independent evaluation (LOSO)

Replacing 5-fold CV with leave-one-speaker-out across 42 pairs reveals: (i) speaker-rich datasets retain accuracy:

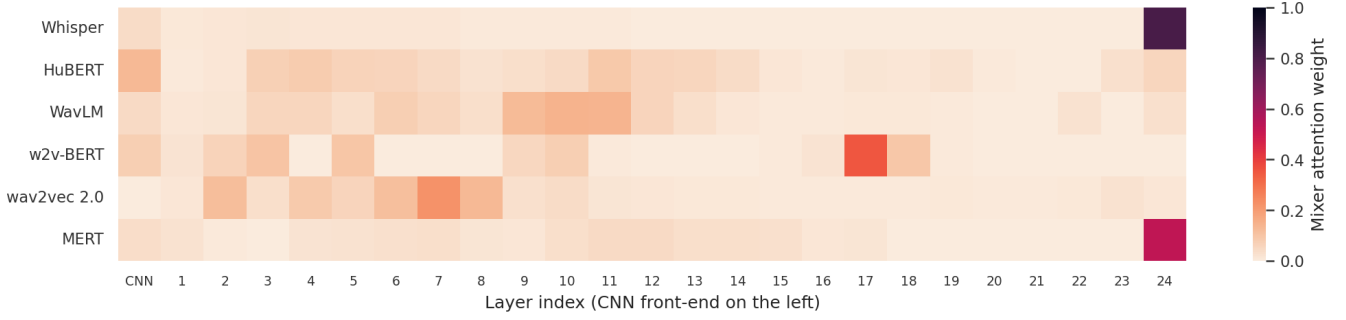


Figure 6: Layer-mixer softmax attention weights, averaged across datasets. Whisper concentrates 64% to 95% of its attention on L24 across all datasets; the SSL encoders spread attention across layers 4 to 13. MERT also collapses heavily to L24 on average.

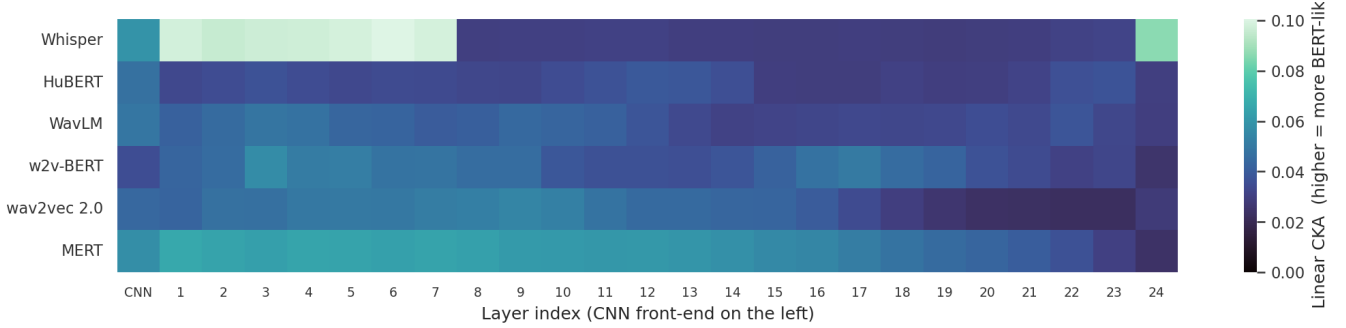


Figure 7: Linear-CKA alignment between each encoder layer and BERT base embeddings of the same utterances, averaged across the seven evaluation datasets. Whisper’s early encoder layers (1 to 7) are most BERT-like; SSL encoders spread alignment across the middle band.

HuBERT/EmoDB drops 0.021 (0.989→0.968 at L11), wav2vec 2.0/EmoDB 0.019, HuBERT/IEMOCAP 0.026; (ii) speaker-poor datasets collapse: MESD averages 0.30 drop, RAVDESS 0.18, worst MERT/MESD 0.32; (iii) the best layer often shifts *deeper* under LOSO (Whisper/SAVEE L9→L24 with $\Delta = +15$, HuBERT/SAVEE L7→L17, WavLM/SAVEE L6→L18), so deep layers are more speaker-invariant on small acted corpora. MSP-Podcast is reported separately because all six LOSO probes are only +0.04 to +0.06 above the 0.507 majority-class baseline.

5.8 Linguistic-bias falsification

Three complementary tests target the central question of whether the layer-wise probe is leaning on lexical structure.

(a) **Congruent vs. incongruent EMIS.** We train a 4-class probe on $\text{EMIS}_{\text{congruent}}$ clips (text emotion = audio emotion) and test on $\text{EMIS}_{\text{incongruent}}$ clips (text $\neq$ audio). We then report $\text{text bias} = \text{proxy_acc} - \text{target_acc}$, where *proxy* predicts the text emotion and *target* takes the audio emotion; positive values indicate the probe took a lexical shortcut. On the explicit slice (Table 6), HuBERT (L20), wav2vec 2.0 (L15), and WavLM (L20) all show large

positive bias (+0.61 to +0.72); **Whisper L24 inverts to -0.19** , the only encoder remaining audio-grounded. The finding is consistent across the all/explicit/implicit slices in `results/TASK7/`.

Table 6: EMIS text bias on the explicit slice (peak layer per encoder). Positive = probe predicts the text emotion. Source: `results/TASK7/probing.*explicit.csv`.

Encoder	Layer	target acc	text bias
HuBERT	L20	0.107	+0.717
wav2vec 2.0	L15	0.123	+0.665
WavLM	L20	0.146	+0.613
Whisper	L24	0.524	−0.191

(b) **Annotator-disagreement ceiling on IEMOCAP.** We split IEMOCAP into 1,308 unanimously-labeled clips and 4,152 clips with ≥ 1 annotator disagreement, and probe each subset (`results/TASK7/`). All six encoders show a high-vs-low-agreement gap of +0.145 to +0.191 at the peak layer (mean +0.169), confirming that the 65 to 70%

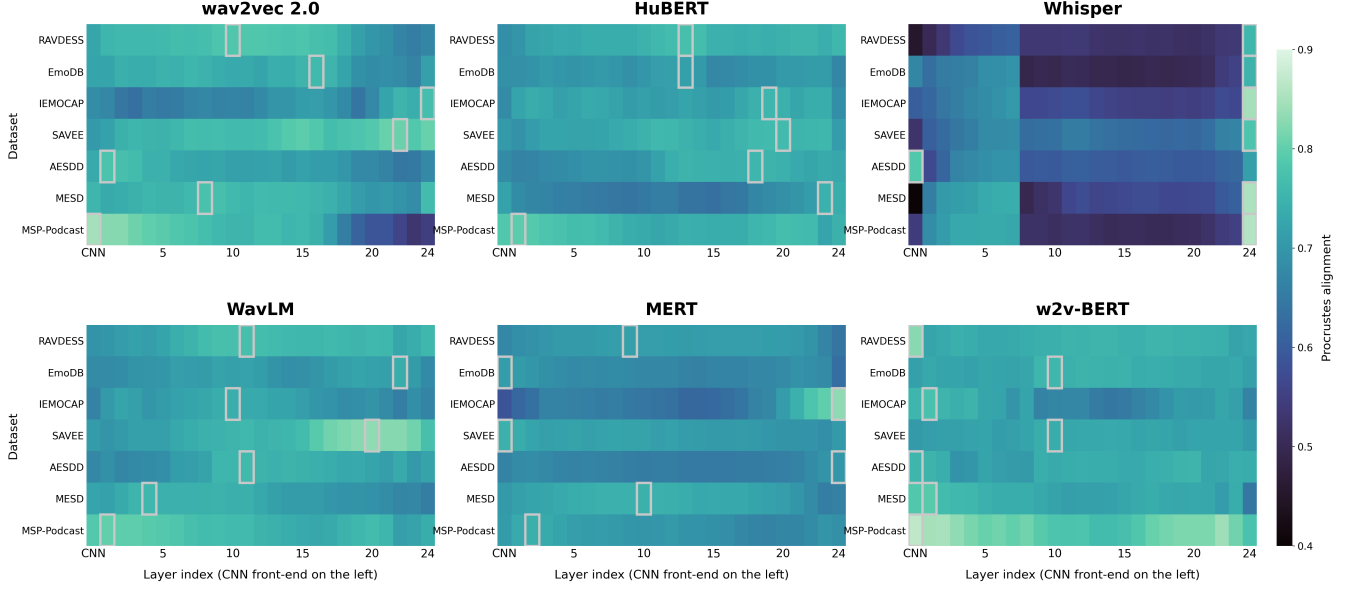


Figure 8: Procrustes alignment between each encoder layer and BERT base embeddings, averaged across the seven evaluation datasets. Whisper’s alignment grows monotonically toward the deepest encoder layer (L24), while SSL encoders peak in the middle band; this complements the CKA pattern in Figure 7 and shows the deep-layer text fingerprint of Whisper most clearly.

IEMOCAP ceiling reported in the midterm reflects label noise rather than a probe limitation.

(c) Adversarial removal (GRL) and CCA projection. On EMIS we attempt to remove the text shortcut at inference time. A two-regime baseline reveals it most clearly: when probes are trained on the RAV+SAV emotion data (no text/emotion correlation), text bias is negative for all four encoders (HuBERT L20: -0.246 ; wav2vec 2.0 L14: -0.086 ; WavLM L20: -0.268 ; Whisper L24: -0.251). When the same probes are trained on EMIS_{congruent} (where text and emotion correlate), three of the four flip strongly positive (HuBERT $+0.928$, wav2vec 2.0 $+0.926$, WavLM $+0.850$); *Whisper alone resists* ($+0.003$). Both attempted post-hoc fixes leave only small Δ bias at safe operating points: GRL ($\lambda \leq 1$) gives at most -0.082 to -0.113 ; CCA projection of the top- k BERT-correlated directions gives at most -0.018 . The shortcut is therefore robust to standard linear and adversarial debiasing on top of a frozen encoder.

5.9 ESC-50 noise robustness

We mix MSP-Podcast clips with ESC-50 environmental sounds at four SNR levels (20, 10, 5, 0 dB) and re-run layer-wise probes (Drive: TASK 1/2/8 noise_analysis_report). The clean-vs-0 dB best-layer accuracy drop varies sharply across encoders (Table 7): Whisper degrades least (mean drop 10.89% at 0 dB; clean 0.864 $\rightarrow$ 0 dB 0.768), while the SSL encoders lose more than twice as much (HuBERT 22.81%, WavLM 27.70%, wav2vec 2.0 29.63%, MERT

28.65%). w2v-BERT sits in between (16.91%). Two consequences: (i) Whisper’s supervised, multilingual, large-scale weak supervision [5] produces representations that survive realistic background noise the best of any encoder we tested; (ii) the optimal layer also drifts slightly deeper as SNR worsens for most encoders (e.g. HuBERT clean L8.6 $\rightarrow$ 0 dB L12.4 on average), suggesting noise pushes the probe to rely on representations that are temporally smoother and less local.

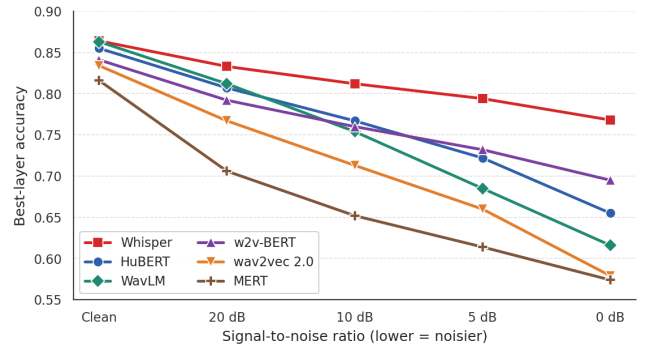


Figure 9: ESC-50 noise robustness on MSP-Podcast. Whisper degrades least; the SSL encoders lose 23 to 30% of clean accuracy at 0 dB SNR.

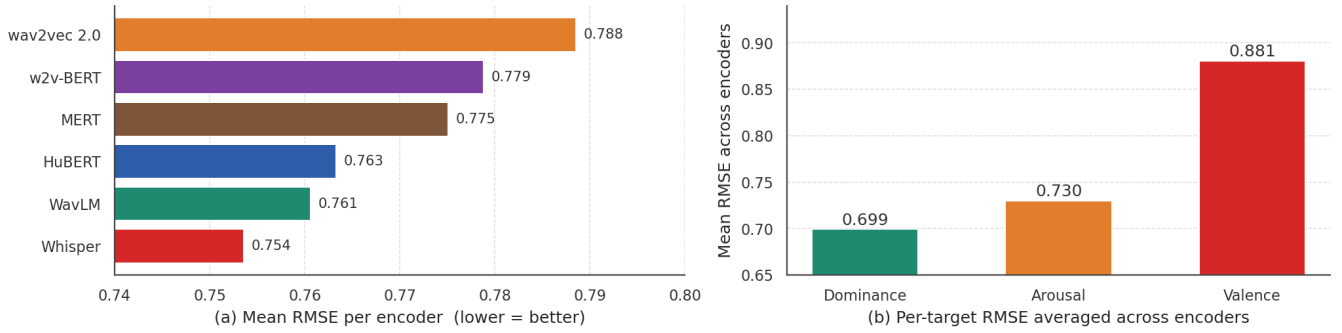


Figure 10: V/A/D regression on MSP-Podcast (ridge probes per layer per dimension). (a) Mean RMSE per encoder; Whisper is best, wav2vec 2.0 worst. (b) Averaged across encoders, valence is by far the hardest dimension, which is the same dimension prior work attributes to lexical content.

Table 7: ESC-50 noise robustness on MSP-Podcast: best-layer accuracy at clean and 0dB SNR. Mean drop is the report’s averaged-across-folds drop, not the simple difference of the two columns. Source: Drive TASK 1/2/8 noisy-summary/noise_analysis_report.txt.

Encoder	Clean	0dB	Δ	Mean drop
Whisper	0.864	0.768	-0.096	10.9%
w2v-BERT	0.841	0.695	-0.146	16.9%
HuBERT	0.855	0.655	-0.200	22.8%
WavLM	0.863	0.616	-0.247	27.7%
MERT	0.816	0.574	-0.242	28.7%
wav2vec 2.0	0.834	0.579	-0.255	29.6%

5.10 Dimensional V/A/D regression

On MSP-Podcast we replace the 4-class probe with three layer-wise ridge regressors targeting valence (EmoVal), arousal (EmoAct), and dominance (EmoDom), in clean and four ESC-50-mixed conditions, and additionally use the predicted V/A/D scores as features for a 4-class emotion classifier. The full results are in TASK 10/11/12 `statistical_analysis.txt`; the three findings the data supports are below.

(i) **Valence is by far the hardest dimension; arousal and dominance are tightly grouped (TASK 10).** Averaged across the six encoders, mean RMSE is 0.881 for valence vs. 0.730 for arousal and 0.699 for dominance — a 26% gap between the easiest and hardest dimension. This is precisely the pattern Wagner et al. [1] and Triantafyllopoulos et al. [2] attribute to valence’s reliance on lexical content; the EMIS test in §5.8 shows the SSL probes do exploit a lexical shortcut on this same dimension at deep layers. The dimension that requires words to predict well is the dimension where the layer-wise emotion code is weakest. Best-layer location reinforces the picture: every encoder’s best dominance layer sits at moderate depth (HuBERT L3, WavLM L3, wav2vec 2.0 L5, MERT L8, Whisper L12, w2v-BERT

L17), shallower than the categorical-emotion peaks in §5.1 (typically L9 to L20), consistent with arousal/dominance being more directly carried by low-level prosodic features than by higher-order phonetic structure.

(ii) **Whisper is best on regression, wav2vec 2.0 worst — same ordering as the noise leaderboard (TASK 11).** Cross-target mean RMSE: Whisper 0.754, WavLM 0.761, HuBERT 0.763, MERT 0.775, w2v-BERT 0.779, wav2vec 2.0 0.789. The same ranking carries through under noise: $\Delta \text{RMSE}_{0\text{dB}-\text{clean}}$ is +0.028 for Whisper, +0.042 for w2v-BERT, +0.060 to +0.062 for HuBERT/WavLM/MERT, and +0.071 for wav2vec 2.0 (Figure 14). This ordering matches the categorical noise ranking in Table 7 encoder-for-encoder; two independent measurement regimes (continuous RMSE on V/A/D, categorical accuracy on 4-class) agree, which is itself evidence the encoder ranking reflects representational quality rather than a quirk of one task setup. Whisper’s advantage on the linguistically-dominated dimension (valence) is consistent with its supervised multilingual pretraining keeping more BERT-like structure at depth (§5.6); the same property that hurts on EMIS (where SSLs go lexical) helps on valence regression (where lexical structure is genuinely informative).

(iii) **V/A/D as features confirms valence is the dominant signal in 4-class emotion (TASK 12).** On all 67,929 MSP-Podcast clips we use predicted V/A/D plus gender as a 4-feature vector to predict the 4-class label (A/H/N/S). Valence’s mutual information with the label is 0.341 (normalized 1.00), more than $1.7\times$ that of arousal (0.197, norm. 0.58) and dominance (0.200, norm. 0.59), and over $20\times$ gender (0.015, norm. 0.04). The Random-Forest macro-F1 is 0.65 (accuracy 0.73) on these four scalar features alone, while logistic regression collapses on the minority class (sad recall 0.06) — symmetric with point (i): the dimension that is hardest to *predict* is also the one most informative once predicted, because the 4-class space is essentially valence-organized (anger and sad share negative valence, happy positive, neutral mid). Full per-classifier breakdown is in Appendix E.

5.11 Task 9 extension: EN-ZH cross-lingual generalization

The Task 9 LOLO results in §5.3 cover four typologically related European languages. To stress-test against a typologically distant one, we add a Mandarin partition and run three conditions on HuBERT, WavLM, and Whisper (Table 5, Figure 4; wav2vec 2.0 was not re-run on the ZH partition). Pooling EN with ZH drops accuracy from the 4-language ceiling to 0.485 to 0.523 across the three encoders. Bidirectional transfer is the strictest test: train EN test ZH peaks at 0.293 (HuBERT, L8); train ZH test EN peaks slightly higher at 0.317 (Whisper, L11). All six transfer numbers sit between 0.28 and 0.32, only 0.03 to 0.07 above the 4-class chance baseline of 0.25. The layer-wise emotion code that pools cleanly across English, German, Greek, and Spanish does not transfer to Mandarin in either direction; the cross-lingual symmetry observed in §5.3 is bounded by linguistic family.

6 Conclusions

Across six encoders, eight datasets, five languages, and the falsification battery above, three claims survive every test we ran:

(C1) Emotion lives in middle-to-late transformer layers. Both midterm probes (3 encoders) and the new probes (3 additional encoders, including the 24-layer architectures) place the emotion peak in the same depth band on every acted dataset, and at the very early layers on spontaneous IEMOCAP. The CREMA-D fixed-lexicon corpus reproduces this depth pattern even when no lexical signal is available.

(C2) The layer-wise signal generalizes, but not uniformly. On the SpeechCraft Task 9 balanced subset, pooled accuracy saturates near 0.94 across encoders; on the full-scale benchmark pooled accuracy retreats to 0.72 to 0.78 and the encoder gap widens, with Whisper leading. LOLO transfer within Europe is asymmetric (German held out generalizes; English held out does not), and bidirectional EN↔ZH transfer collapses to within 0.07 of chance. Speaker leakage is small on speaker-rich corpora (≤ 0.03 drop) but severe on small ones (MESD approximately 0.30). LoRA fine-tuning at the best layer hurts on 29 of 30 combinations, indicating the relevant features are already present in the frozen encoder.

(C3) Self-supervised encoders take a lexical shortcut when one is offered; Whisper does not, and is also the most noise-robust. The EMIS incongruent test gives HuBERT, wav2vec 2.0, and WavLM text-bias values of +0.61 to +0.72 at their deep layers, while Whisper L24 inverts to -0.19. The two-regime baseline confirms this: three SSL encoders flip from negative to large positive bias when the training set offers a text shortcut; Whisper does not. BERT alignment and the layer mixer agree: Whisper concentrates attention and BERT alignment at L24, the same depth where the EMIS test shows it remains audio-grounded.

Whisper additionally dominates the noise-robustness leaderboard (approximately 11% drop at 0 dB vs. approximately 23 to 30% for the SSL encoders) and the dimensional-regression noise-resilience ranking ($\Delta\text{RMSE} = +0.028$ vs. $+0.04$ to $+0.07$). On the dimension that depends most on words (valence), every encoder still does worst, exactly as prior work predicts. Standard adversarial and projection-based debiasing do not remove the SSL shortcut at any safe operating point. Finally, while the European-language LOLO transfer is asymmetric but nonzero, replacing the held-out language with Mandarin Chinese (Task 9) collapses pooled HuBERT to weighted F1 0.485 and EN-trained-ZH-tested to 0.293; the layer-wise emotion code that transfers across English, German, Greek, and Spanish does not transfer outside that family.

The actionable take-away is narrow but robust: probes that report layer-wise emotion accuracy on small acted corpora should be paired with at least one falsification test (lexicon control, EMIS, noise) before the result is interpreted as “the model hears emotion in the voice”. The same probe accuracy can be reached two different ways, and our experiments separate them.

Limitations

There are three places where we want to be honest about the limits of what these results show. First, all six encoders we tested were pretrained on data that is heavily English weighted. Whisper is multilingual but its training set is English dominant, and the SSL encoders are even more so. The asymmetric LOLO pattern (German held out generalizes well, Spanish held out collapses) and the sharp Mandarin transfer drop in §5.11 therefore reflect pretraining data distribution as much as anything intrinsic to the architectures or the pretraining objectives. Second, our LoRA experiments tune at the per-combination best probing layer rather than across the entire encoder. The headline that frozen wins on 29 of 30 combinations specifically means LoRA at one layer cannot beat the right frozen layer; full-encoder fine-tuning was outside our compute budget and is a separate question. Third, our linguistic-bias falsification battery shows the SSL text shortcut survives standard adversarial and projection-based debiasing, but that is not the same as proving no debiasing technique could remove it. Stronger representation surgery methods like concept erasure applied to deep SSL layers might tell a different story; we did not have time to test that.

7 Team Contributions

Nima Kelidari. Built the Slurm-based experiment pipeline; extended the probing grid to six encoders by adding WavLM, w2v-BERT, and MERT; built and ran the ESC-50 noise-augmentation pipeline at four SNR levels (§5.9, Drive: TASK 1/2/8) and the MSP-Podcast spontaneous-speech pipeline; led the V/A/D dimensional-regression track end-to-end: clean, noisy, and V/A/D-as-features for classification (§5.10, Drive: TASK 10/11/12). Co-authored

the proposal and led the literature review; coordinated report formatting, figure/table styling, cross-references, and document consistency; ran on the order of 360 frozen-layer experiments.

Minoo Ahmadi. Built LOSO evaluation on the frozen probing pipeline across selected layers (§5.7). Gained access to the EMIS dataset and implemented a text-bias diagnostic using sentence-stratified probing on synthetic congruent clips, evaluated on incongruent clips (§5.8). Developed an IEMOCAP annotator-disagreement diagnostic by grouping clips by consensus and comparing accuracy across subsets. Co-authored the project proposals; produced the GRL adversarial and CCA-projection debiasing experiments and the corresponding summary plots and READMEs (artifacts in `results/LOSO/`, `results/TASK6/`, `results/TASK7/`).

Chaitanya Parwatkar. Built the original layer-wise probing pipeline (hidden-state extraction, mean pooling, `StandardScaler` + 5-fold logistic regression); ran the BERT alignment experiments via CCA, CKA, Procrustes, and centroid transfer to measure how text-like each layer is (`results/bert_alignment/`); built the learned layer mixer with softmax attention over 25 hidden states; ran LoRA across the 30 (encoder, dataset) combinations; added CREMA-D probing as the cleanest fixed-lexicon test (`results/cremad/`). Wrote the original Method, Results, and Cross-Model Correlation sections of the midterm.

Xiangxu Lin. Compiled the SpeechCraft Task 9 benchmark across the six encoders for multilingual pooled and leave-one-language-out cross-lingual stress tests (balanced subset, §5.3); ran the full-scale SpeechCraft pooled benchmark on four encoders and the EN-ZH partition (5-language pooled and bidirectional EN↔ZH transfer) on HuBERT, WavLM, and Whisper (§5.11, Table 5, Figure 4). Initiated the midterm report and wrote the original Data Description section.

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A Per-class counts (six base datasets, CREMA-D, Task 9)

The original six datasets follow the midterm Appendix A counts: RAVDESS (192 neutral; 384 each for calm, angry, disgust, fearful, happy, sad, surprised), EmoDB (127/81/79/71/69/62/46 across angry, bored, neutral, happy, fearful, sad, disgust), IEMOCAP (1,708 neutral, 1,636 happy, 1,103 angry, 1,084 sad), SAVEE (120 neutral; 60 each for the remaining six classes), AESDD (approximately 121 per emotion), and MESD (approximately 144 or 143 per emotion). CREMA-D: 7,442 utterances from 91 actors saying the same 12 sentences across 6 emotions (anger, disgust, fear, happy, neutral, sad); per-actor accuracies for our best probe are tabulated in `results/cremad/cremad_per_actor.csv`, per-sentence in `results/cremad/cremad_per_sentence.csv`. SpeechCraft Task 9 balanced subset (4-class manifest): English 2,546, German 375, Greek 483, Spanish 576; class counts: happy 831, sad 1,768, fear 701, disgust 680.

B Extended Task 9 (SpeechCraft) figures

The figures below complement Table 4 and Figure 3 in the main body, showing per-encoder pooled accuracy curves with peak layers marked, the LOLO heatmap, and the pooled-minus-LOLO gap.

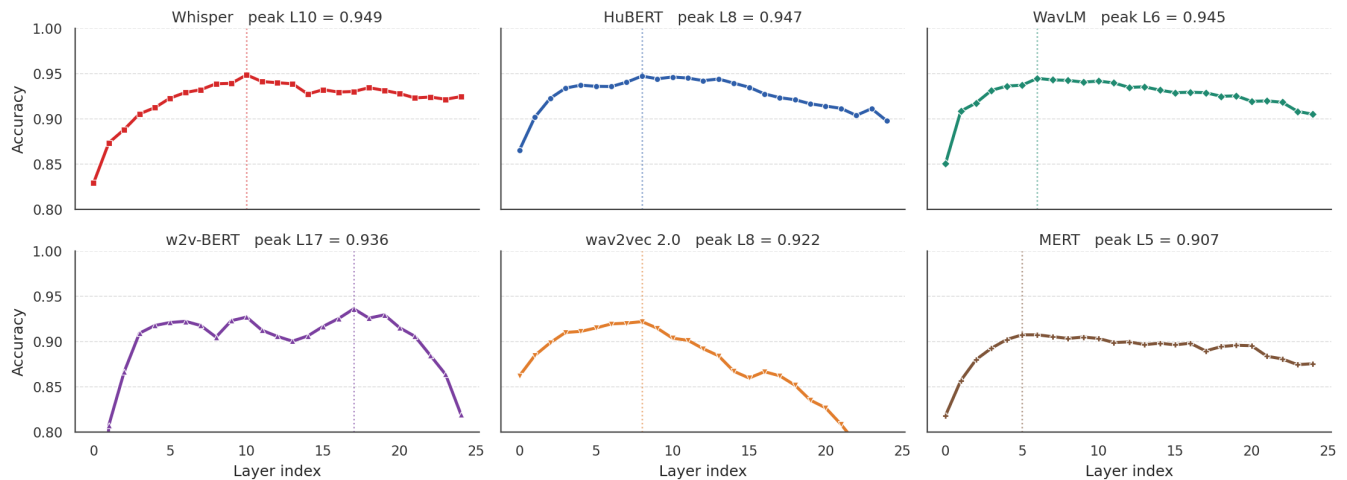


Figure 11: Task 9 (SpeechCraft) pooled per-encoder layer-wise accuracy curves; vertical dotted line marks each encoder’s peak layer.

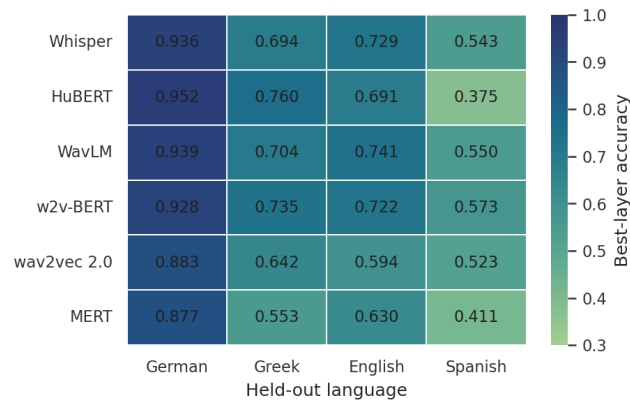


Figure 12: Task 9 LOLO accuracy heatmap (encoders × held-out language). German held out is uniformly easier than Spanish or English held out.

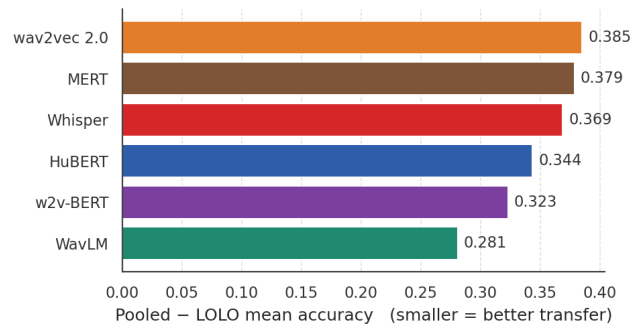


Figure 13: Pooled-minus-LOLO accuracy gap per encoder; smaller is better.

Table 8: Best-layer accuracy on MSP-Podcast across the six encoders and four noise levels. Source: Drive TASK 1/2/8 noise_analysis_report.txt.

Encoder	Clean	20 dB	10 dB	5 dB	0 dB
HuBERT	0.855	0.807	0.767	0.722	0.655
MERT	0.816	0.706	0.652	0.614	0.574
WavLM	0.863	0.812	0.754	0.685	0.616
Whisper	0.864	0.833	0.812	0.794	0.768
w2v-BERT	0.841	0.792	0.760	0.732	0.695
wav2vec 2.0	0.834	0.767	0.713	0.660	0.579

Table 9: Per-target RMSE averaged across the six encoders, on MSP-Podcast at five noise levels. Source: Drive TASK 10/11 statistical_analysis.txt.

Target	Clean	20 dB	10 dB	5 dB	0 dB
EmoDom (dominance)	0.699	0.710	0.718	0.725	0.735
EmoAct (arousal)	0.730	0.746	0.759	0.770	0.786
EmoVal (valence)	0.881	0.897	0.908	0.917	0.929

C Per-encoder, per-SNR best-layer accuracy (ESC-50 noise)

D V/A/D regression details

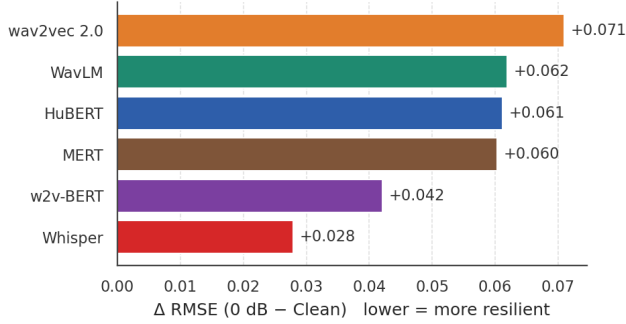


Figure 14: V/A/D regression noise resilience: $\Delta\text{RMSE}_{0\text{dB}-\text{clean}}$ per encoder. Whisper is the most resilient.

Table 10: Best-layer RMSE per encoder for each V/A/D target under heavy 0dB noise. “L” = encoder layer index. EmoVal best layers cluster near the deep end for noise-tolerant encoders (Whisper L24, HuBERT L21, WavLM L20). Source: Drive TASK 11 statistical_analysis.txt.

Encoder	EmoAct		EmoDom		EmoVal	
	L	RMSE	L	RMSE	L	RMSE
Whisper	13	0.732	22	0.701	24	0.837
w2v-BERT	17	0.758	17	0.722	13	0.899
HuBERT	10	0.772	10	0.726	21	0.908
WavLM	9	0.769	9	0.726	20	0.895
MERT	10	0.774	11	0.723	9	0.944
wav2vec 2.0	8	0.792	7	0.739	9	0.937

E V/A/D-as-features for emotion classification (Task 12)

Treating valence, arousal, dominance, and gender as features for a 4-class emotion classifier (TASK 12 `statistical_analysis.txt`, $n=67,929$ MSP-Podcast clips) confirms valence dominates importance: mutual information 0.341 (normalized 1.00) for valence, 0.197 for arousal (norm. 0.58), 0.200 for dominance (0.59), and 0.015 for gender (0.04). The RF-importance ranking matches the MI ranking. Three classifiers were compared on the same features (Table 11): all three reach ≈ 0.73 accuracy, but the minority sad class is the failure mode — logistic regression collapses to recall 0.06, Random Forest recovers to 0.25 by leveraging interactions, and the MLP sits between the two. Anger is the easiest class for every classifier (recall 0.72 to 0.79), consistent with the V/A/D representation cleanly separating high-arousal negative-valence cases.

Table 11: V/A/D-as-features 4-class classifier comparison (MSP-Podcast, $n=67,929$). “A” anger, “H” happy, “N” neutral, “S” sad. Per-class recall shows the sad-collapse pattern.

Classifier	Acc	Macro F1	R(A)	R(H)	R(N)	R(S)
Logistic Reg.	0.727	0.585	0.72	0.59	0.87	0.06
Random Forest	0.734	0.646	0.75	0.55	0.89	0.25
MLP Classifier	0.735	0.637	0.79	0.57	0.87	0.21

F Code, data, and reproducibility

All code, manifests, per-(model, dataset) CSVs, and notebooks are in the project repository at github.com/Nik-elroid/linguistic-agnostic-ser. Key branches: `main` (probing pipeline, CREMA-D, layer mixer, LoRA, BERT alignment), `minoo` (LOSO, GRL/CCA, EMIS, IEMOCAP-disagreement), `speechcraft` (Task 9 balanced subset). The W&B project AGSER/linguistic-agnostic-ser contains all run logs.